

Projections of Heat-Stress Conditions Across Bangladesh Using Explainable Machine Learning and Climate Trend Integration

Mohammad Khalid Hasan Nabil^{1*}, Tahia Parsha¹,
Musayeb Hossain Usama¹, Atika Akter¹, Ivan Grgić²

¹*Department of Information Communication Technology, Faculty of Science and Technology, Bangladesh University of Professionals, Dhaka, 1216, Bangladesh.

²Mechanical Engineering Faculty in Slavonski Brod, University of Slavonski Brod, Slavonski Brod, 35000, Croatia.

*Corresponding author(s). E-mail(s):
nabil23549009015@student.bup.edu.bd;

Abstract

This is the title page for the manuscript “Projections of Heat-Stress Conditions Across Bangladesh Using Explainable Machine Learning and Climate Trend Integration”. It carries the author, affiliation and declaration information that is withheld from the anonymised manuscript file.

Corresponding author

Mohammad Khalid Hasan Nabil. ORCID: 0009-0007-3255-1420. Email:
nabil23549009015@student.bup.edu.bd

CRedit authorship contribution statement

Mohammad Khalid Hasan Nabil: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Project administration.

Tahia Parsha: Investigation, Software, Writing – original draft, Writing – review &

editing.

Musayeb Hossain Usama: Investigation, Software, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The raw daily meteorological data are available from the Visual Crossing Weather API (<https://www.visualcrossing.com>). The complete deterministic machine-learning pipeline source code, the interactive Streamlit web-application dashboard, and the comprehensive 64-district projected dataset are permanently archived on Zenodo (<https://doi.org/10.5281/zenodo.21628776>). The interactive dashboard, which visualises the projections, validation, and risk categories for all 64 districts and seven conditions, is served live at

(<https://tcc-64-districts-mohammad-khalid.streamlit.app/>).